



USER GUIDE

KusumaVision NMS Complete Guide

Managing ZTE, C-Data, and HiOSO OLTs — from installation, monitoring, and ONU registration to alarms, maps, and ODPs.

EDITION

Open Source

LANGUAGE

English

DATE

1 October 2026

APPLICATION

NMS v2

KusumaVision NMS — open-source software under the MIT licence.

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01 Introduction

About this guide

This guide explains how to use KusumaVision NMS day to day: adding OLTs, monitoring ONUs, registering new subscribers, handling alarms, and managing the map and ODPs. Every feature follows the same pattern: where it is, what it does, who may use it, the steps, and its limits per vendor.

This is the **public edition** for users of the open-source release (MIT licence). Chapter 2 explains installation on your own server; the other chapters cover daily use.

Conventions:

- **Labels like this** are menu, button, or tab names exactly as shown on screen (English interface).
- **text like this** is a CLI command, an input value, or an address.
- **Caution** and **Danger** boxes mark actions that can interrupt subscriber service. Read them before pressing the button.

What KusumaVision NMS is

KusumaVision NMS is a web-based network management system for FTTH (GPON/EPON). It works much like SmartOLT or NetNumen: it monitors OLTs and ONUs centrally, registers new ONUs, changes configuration over CLI, and sends alarms to Telegram and the Android app.

How the NMS talks to OLTs:

- **SNMP (read)**: port status, ONU lists, online status, and RX power are read over SNMP v1/v2c at regular intervals (polling).
- **Telnet (write)**: ONU registration, reboots, configuration changes, and saving the configuration are sent over a Telnet session to the OLT.
- **Cache**: the last polling result is stored so pages open quickly without contacting the OLT every time. Figures on screen come from the **last poll**, not this second, unless a page clearly reads live (for example ONU Detail).

Supported vendors and models

FAMILY	TESTED MODELS	COVERAGE
ZTE GPON	C300, C320	Most complete: monitoring, ONU registration, ONU configuration, bulk TR069, configuration backup, PON port disable/enable
ZTE GPON	C600 (TITAN)	Monitoring, TR069-model ONU registration, ONU delete/reboot; some write actions are not available yet
C-Data EPON	FD1304E (V3 firmware); FD1108/FD1208/FD1504 series	Monitoring, ONU actions (rename, reboot, enable/disable, delete), VLANs on V3 firmware
C-Data GPON	FD1608S (FlashV3), FD1601S, FD1602S	Monitoring, ONU actions, Remote ONT (FlashV3), VLANs and port detail (V3 firmware)
HiOSO EPON	HA7304 (4 ports), HA7302 (2 ports)	Monitoring, ONU actions (rename, reboot, enable/disable, delete)

NOTE

V-Sol OLTs are not supported yet. HsAirPo/HSGQ support has been removed. OLTs outside this list may be detected incorrectly or have no features at all.

ONU registration (provisioning), ONU configuration, bulk TR069, and configuration backup are currently **ZTE only**.

Key terms

TERM	MEANING
OLT	The central device on the ISP side that serves many ONUs through PON ports
PON port	An optical OLT port; one GPON port serves up to 128 ONUs
ONU / ONT	The device at the subscriber's home
SN (serial number)	GPON ONU identity, for example <code>ZTEGC0A1B2C3</code> . EPON ONUs use their MAC address
RX / attenuation	Optical power received by the ONU (dBm). Normal is roughly -15 to -25 dBm
ODP	A field distribution point/splitter where subscriber drop cables branch off
Polling	Periodic SNMP reading by the NMS (every 5 minutes by default)
Unconfigured	An ONU that is connected by fibre but not yet registered on the OLT
TR-069 / ACS	Protocol and server for managing ONUs remotely (WiFi, firmware)
<code>write</code>	OLT command that saves the running configuration to permanent memory

User roles

ROLE	FOR	SUMMARY
Admin	System owners	All features, including Settings, Audit Logs, and user management
Operator	NOC / technicians	All operational actions on the OLTs they can access; no Settings or Audit Logs
Partner	Partner ISPs	Only OLTs assigned to them plus their own OLTs; has their own Telegram bot
Demo	Prospects	Sees sample data only; every change is rejected

Per-feature permissions are listed in the Users and Permissions chapter.

02 Getting Started

Requirements

- An Ubuntu 22.04/24.04 server (at least 2 GB RAM) for a direct installation, or a Windows/macOS/Linux PC with Docker (at least 4 GB RAM).
- Network reachability from the server to the OLTs: SNMP `UDP/161` and Telnet `TCP/23`.
- Components installed by the installer: PHP 8.3, PostgreSQL, Redis, Nginx, Supervisor, Go (for the SNMP poller), Net-SNMP, and Node.js (to build the interface).

Readiness checker: `bash scripts/check-requirements.sh` (add `--lang en` for English output).

Installing with Docker

Suited to a single office PC, much like a NetNumen installation.

1. Copy the example file: `cp .env.docker.example .env`.
2. Change at least `DB_PASSWORD` and the admin details `ADMIN_NAME`, `ADMIN_EMAIL`, `ADMIN_PASSWORD`. Optional: `APP_PORT` (default 8080), `APP_URL`, `ACS_URL` / `ACS_USERNAME` / `ACS_PASSWORD`, `TRUSTED_PROXIES`.
3. Run `docker compose up -d --build`. On Windows, double-click `start.bat` (there are also `stop.bat` and `update.bat`).
4. Open `http://localhost:8080` and sign in with the admin account from step 2.

The APP_KEY is generated automatically, database migrations run automatically, and the admin account is created while the users table is still empty. Back up the database with `docker compose exec -T db pg_dump ...`.

Installing directly with install.sh

1. Clone the repository to `/var/www/KusumaVisionNMS`.
2. Run `sudo bash install.sh` (add `--lang en` for an English installer, `--yes` for non-interactive mode).
3. Answer the installer: language, `APP_URL` (default `http://<server IP>`), database name/user/password (blank = generated), and the admin account (password at least 8 characters).
4. The installer installs every component, writes `.env`, builds the Go SNMP poller, runs migrations, configures Nginx, and starts three Supervisor daemons: `kusumavision-worker` (queue), `kusumavision-scheduler` (polling, alarms, backups), and `kusumavision-telnet-proxy` (browser Telnet terminal).
5. At the end the installer prints a summary with the database password and admin account. **Keep it safe.**
6. Add HTTPS with `certbot --nginx -d nms.your-domain.com`.

Important .env settings

KEY	PURPOSE
<code>APP_URL</code>	Full application address, for example <code>https://nms.your-domain.com</code>
<code>SNMP_POLLER_DRIVER</code>	<code>go</code> (recommended, fast) or <code>php</code>
<code>TRUSTED_PROXIES</code>	Required behind Cloudflare or a load balancer; without it sign-in fails with error 419
<code>SESSION_SECURE_COOKIE</code>	<code>true</code> when serving over HTTPS
<code>ACS_URL</code> , <code>ACS_USERNAME</code> , <code>ACS_PASSWORD</code>	TR-069 ACS server for the TR069 features (can also be set in Settings)
<code>FIREBASE_CREDENTIALS</code>	Firebase credentials for Android app push notifications

After every `.env` change, run `php artisan config:cache`, then `php artisan queue:restart` and `sudo supervisorctl restart kusumavision-telnet-proxy`.

Creating additional admin accounts

Command: `php artisan user:create --name="Name" --email=admin@domain.com --password=secret123 --role=admin` (on Docker, prefix it with `docker compose exec app`). Without `--role=admin` the account is created as an operator.

Updating the application

- Docker: `git pull && docker compose up -d --build` (or `update.bat`).
- Direct installation: `git pull`, `composer install --no-dev --optimize-autoloader`, `npm ci && npm run build`, rebuild the Go poller (`go build -o bin/kv-snmp-poller ./cmd/kv-snmp-poller`), `php artisan migrate --force`, `php artisan optimize`, `php artisan queue:restart`, and restart `kusumavision-telnet-proxy`.

After an update, ask users to reload with Ctrl+Shift+R.

Signing in and out

1. Open the application address and click `Login`.
2. Enter your email and password. Tick "Remember me on this device" if needed.
3. Click `Sign In`.

Self-registration is deliberately disabled: accounts are created by an admin. The "Forgot password?" link sends a reset email; it requires SMTP configured in `.env` (the default `MAIL_MAILER=log` only writes emails to the log). Sign out from the account menu (avatar, top right) → `Log Out`.

Application layout

The **left sidebar** is grouped as follows:

GROUP	MENU
Overview	Dashboard
OLT network	SmartOLT , Unconfigured , ONU Monitoring
Field	ONU Map , ODP
Monitoring	Alarms , Reports
Administration	Users (admin), My Telegram Bot (partner), Audit Logs and Settings (admin)
Help	Guide

The sidebar can be collapsed with the arrow on its edge; the browser remembers your choice. On phones, open it from the menu icon. The **System** panel at the bottom of the sidebar (desktop) shows the version, server time, uptime, and number of users online.

The **top header** contains:

- The global search box ([Search devices, OLTs, ONUs, or locations...](#) , shortcut ⌘K or Ctrl+K).
- The [Language](#) picker, the [Notifications](#) bell, and the account menu ([Profile](#) , [Appearance](#) , [Log Out](#)).

Language and theme

- **Language:** click [Language](#) in the header and choose Bahasa Indonesia or English. The choice is saved to your account and used on every device.
- **Theme:** account menu → [Appearance](#) → [Dark](#) , [Light](#) , or [Follow System](#) . On phones, the theme options are at the bottom of the side menu.

Profile

The [Profile](#) page has three sections: [Profile Information](#) (name and email), [Update Password](#) , and [Delete Account](#) . You can also change the theme here.

03 Dashboard and Monitoring

Dashboard

The **Dashboard** summarises the whole network you are allowed to see.

- **Four main cards:** **Total OLTs** (online/offline), **Total ONUs**, **ONUs Online** (share of the total), and **Active Alarms** (critical/major counts).
- **Network Activity Trend (Polling):** polling success chart. Choose 24 hours, 7 days, or 30 days. A rising red line means an OLT failed to be read.
- **ONU Status:** donut chart of online, warning, and offline ONUs.
- **OLT Inventory:** OLTs per model with Up/Down status; click to open the detail.
- **Latest Alarms:** the five most recent alarms, with a **View All** link.
- **Provisioning Jobs:** ONU registrations that are pending, in progress, successful, and failed.
- **Remote ONU Actions:** find an ONU and run **Reboot**, **Enable**, or **Disable** without opening the port page.

NOTE

The **Reset**, **Upgrade**, and Log buttons on the Remote ONU Actions card do not work yet. Their dialog says the feature is not available.

All figures follow your OLT scope: partners only see their own OLTs and those assigned to them.

Global search

Press ⌘K (Mac) or Ctrl+K, or click the search box in the header.

1. Type at least 2 characters: an OLT name or IP, a serial number, a subscriber name, an interface (for example **1/2/3:15**), or an ONU MAC.
2. Pick a result with ↑/↓ and press Enter. Esc closes the box.
3. An OLT result opens the OLT detail. An ONU result opens the per-port ONU page with that ONU highlighted.

Results are capped at 10 (at most 5 OLTs) and come from the last poll.

TIP

Although the placeholder mentions "locations", this search does not look up subscriber addresses, map pins, or ODPs. Use the **ODP** menu for ODPs.

ONU Monitoring

ONU Monitoring shows the ONUs of one OLT in a single filterable table. It is read-only; ONU actions are done on the port page.

1. Choose an OLT in **Select OLT...**.
2. Narrow the list with the filters:
 - Port, Status (Online, LOS, Dying Gasp, Offline), and Admin (Active, Disabled).

- Attenuation: Normal, Warning, Critical, or No RX Data.
- ODP: **All ODPs**, **No ODP**, or a specific ODP.

3. Summary cards show Total ONUs, Online, LOS / Power Loss, and Offline.

4. Click **Open in port (manage ONU)** on a row to manage that ONU.

Scan this OLT's ONUs rereads every ONU on the OLT over SNMP (seconds to tens of seconds for a large OLT).

Attenuation thresholds used across the application:

LEVEL	RX VALUE
Normal	above -25 dBm and below -10 dBm
Warning	-25 to -28 dBm, or -10 to -8 dBm
Critical	-28 dBm or lower, or -8 dBm or higher

04 Managing OLTs

This chapter applies to every vendor. Vendor-specific features are in the ZTE, C-Data, and HiOSO chapters.

OLT list

SmartOLT shows every OLT you can access, in three tabs: **OLT ZTE**, **OLT C-Data**, and **OLT HiOSO**. Each row shows the OLT name (with a **Private** badge if a partner owns it), driver/family, SNMP address, version, auto-poll status, alarm status, last test result, and last poll time.

Row action icons:

ICON	PURPOSE	WHO
Details	Opens the OLT detail	Everyone
Alarm bell	Turns alarm notifications for this OLT on or off	Admin, operator, partner
Test SNMP	Tests the SNMP connection	See Test SNMP
Edit	Edits the OLT	Admin, operator, partner (connection fields owner only)
Profile	Profile catalogue (ZTE only)	Admin, operator, partner
Save	Saves the running configuration to OLT memory	Admin, operator, partner; needs Telnet
Telnet to OLT	Browser CLI terminal	See Telnet terminal
Delete OLT	Removes the OLT from the NMS	Admin, operator; partners only their own OLTs

Adding an OLT

1. Open **SmartOLT**, choose the vendor tab, and click **Add OLT**.
2. **OLT identity**: fill in the OLT Name, Vendor/Family, and IP Address. The IP and SNMP port combination must be unique.
3. **SNMP configuration**: SNMP Port (default **161**), SNMP Version (**v2c** recommended), SNMP Read Community (required), and SNMP Write Community (optional, see the note below).
4. **CLI configuration**: set CLI Transport to **Telnet**, then fill in the CLI Port (**23**), CLI Username, and CLI Password. Without these, write features (registration, reboot, save configuration, terminal) cannot be used.
5. **SNMP auto-poll**: keep **Enable SNMP auto-poll** on. The Polling Interval and ONU RX Interval default to 5 minutes (range 1–1440).
6. Click **Save OLT**. For C-Data and HiOSO, the NMS scans the ONUs immediately and shows how many it found.
7. Click **Test SNMP** on the new row to confirm the connection.

CAUTION

SNMP v3 and SSH appear in the options but are **not supported yet**. Use SNMP v1/v2c and Telnet.

When the **SNMP Write Community** is required:

- ZTE C300/C320: to rename ONUs and enable/disable ONUs (both use SNMP SET).
- HiOSO HA7302: to rename ONUs.

When editing an OLT, secret fields (communities, passwords) left blank keep their old values.

Partner-owned private OLTs

Partners may add their own OLTs. Such an OLT automatically becomes **Private** to that partner: it is hidden from admins/operators, and its notifications go only to the owner. On OLTs that are merely assigned to them, partners **cannot** change the IP/credentials, test the connection, open Telnet, or read configuration backups.

Test SNMP

The **Test SNMP** icon reads the OLT system information (sysDescr, sysObjectID), detects the vendor and model, and stores the result without clearing ONU data. The **Test** column changes to OK, Failed, or Not tested.

- For ZTE C600, run Test SNMP once after adding the OLT so the NMS recognises it as a C600.
- For C-Data GPON, the test also checks whether the firmware is FlashV3.
- Who may test: admins/operators or the partner who owns the OLT.

Automatic polling and scanning

The NMS scheduler checks every minute and polls OLTs whose auto-poll is on and whose interval has elapsed. Each poll reads ports, ONUs, status, and RX power, then evaluates alarms.

Manual refresh options:

BUTTON	WHERE	SCOPE
Refresh SNMP	ZTE OLT detail, GPON Ports page	Every port and ONU on the OLT
Refresh ONU	Per-port ONU page	One port only
Scan ONUs	C-Data/HiOSO OLT detail	The whole OLT
Scan this OLT's ONUs	ONU Monitoring	The whole OLT

Manual scans are limited to 30 per minute per user. C-Data/HiOSO pages also scan automatically when the data is older than 5 minutes, so they can feel slow when first opened.

Per-OLT alarm switch

The bell icon on an OLT row turns **delivery** of notifications for that OLT on or off. Alarms are still recorded and remain visible on the **Alarms** page.

- Admins and operators control the switch for admin/operator recipients: the main Telegram bot and app push.
- Partners control their own switch (their Telegram bot and push), which does not affect admins.

Use this switch during scheduled maintenance to avoid a flood of notifications.

Save configuration (Save Config)

The save icon on an OLT row sends the save command to the OLT (`write` for ZTE and HiOSO, `save` for C-Data). The running configuration is copied to the OLT's permanent memory.

1. Click the save icon on the OLT row.
2. Read the confirmation and click `Save`. On a ZTE C300 with a large configuration this can take about 30 seconds.

CAUTION

Many NMS actions do **not** save automatically (reboot, ONU delete, C300/C320 registration, port disable, and more). If the OLT reboots before a save, those changes are lost. Conversely, **Save Config also saves states you may not want to keep**, such as a port that is currently disabled. The per-action summary is in the Safe Practice chapter.

Save Config is **not** a backup to the NMS database. Configuration backup (ZTE only) is covered in the ZTE chapter.

Telnet terminal

The `Telnet to OLT` icon opens a terminal window in the browser and signs in to the OLT automatically. The window can be moved, minimised, or maximised.

- Requires: CLI Transport set to Telnet and CLI credentials filled in.
- Who may open it: admins/operators or the partner who owns the OLT.
- Every terminal session is recorded in the `Audit Logs`.

DANGER

The terminal gives full CLI access. Commands such as `no onu`, `shutdown`, or `write` take effect immediately on production equipment.

Deleting an OLT

`Delete OLT` removes the OLT and its cached data from the NMS. The device configuration is not touched. This is permanent. Admins/operators can delete any OLT in their scope; partners only their own.

05 ZTE OLTs (C300, C320, C600)

ZTE has the most complete support. Almost every feature applies to the C300 and C320; the C600 has a few exceptions, summarised at the end of this chapter.

OLT detail and chassis

Click **Details** on a ZTE OLT row.

- **Header buttons:** **Back**, **Edit**, **GPON Port & ONU**, **Registrations** (registration history), **Config Backup**, and **Refresh SNMP** (rescan every port and ONU).
- **Summary cards:** OLT status, number of GPON ports, total ONUs (online/total), last refresh time, and system information (name, description, uptime).
- **Chassis view:** boards with CPU/memory and port LEDs. Click **Refresh Hardware** to read them over CLI. Click a GPON or uplink port (xgei/gei) to open Port Detail.

GPON Ports page

GPON Port & ONU opens a grid of port cards. Each card shows UP/DOWN status (red when down), the port description, and an online/total ONU bar. The search box accepts a port number, subscriber name, MAC, or ONU serial. Click a card to open the ONU list for that port.

Port Detail

Open it by clicking a port in the chassis view. **Refresh from OLT** reads the latest data over CLI.

- **Port Status:** admin status, link, negotiation, registered ONUs and capacity.
- **Optical / SFP:** module RX/TX power (coloured when outside the OLT thresholds), vendor, part number, serial, wavelength, bias, temperature, and voltage.
- **Traffic:** input/output, throughput, and peaks. Uplink ports also offer **Live Traffic** (chart refreshed every 10 seconds).

Editing the PON port description

1. In the Description box, click **Edit**.
2. Enter the text (up to 64 characters), or clear it to remove the description.
3. Click **Save**. The description is written to the OLT over CLI **and saved immediately** (**write**).

VLANs on uplink ports

Uplink ports list their tagged VLANs. To add one, fill in **Add tagged VLAN** (1-4094) and click **Add & Tag VLAN**. The NMS creates the VLAN if needed, tags it on the uplink, and **saves the configuration** (**write**) **automatically**.

Disabling and enabling a PON port

This disconnects or reconnects **every subscriber on one port** at once. Use it for fibre route maintenance, fault isolation, or a port that is intentionally out of service.

- **Where:** the red **Disable Port** or **Enable Port** button in the Port Detail header (GPON ports).

- **Who:** only **admins**, or the **partner who owns** a private OLT. Operators cannot.
- **Models:** C300 and C320. Not yet available on the C600.

Steps:

1. Open the Port Detail and check the number of registered ONUs.
2. Click **Disable Port**. The confirmation dialog shows how many ONUs will be disconnected.
3. Click **Disable Port** in the dialog. The NMS sends **shutdown** to the port interface.
4. One **"PON port disabled by admin"** notification goes to Telegram and the app. The port page shows a yellow banner while the port is down.
5. To bring it back, click **Enable Port** and confirm. The NMS sends **no shutdown** and one "port enabled again" notification. This is a recovery message, so it is only delivered when **Send when an alarm clears** is on under **Settings** → **Alarm** (off by default on a new installation).

While the port is down, port-down, ONU-offline, and ODP alarms for that port are **held back** so they do not flood recipients. Every action is recorded in the **Audit Logs**.

DANGER

This command **does not save automatically**, so the port comes back up if the OLT reboots. However, if someone runs **Save Config**, edits a port description, or adds an uplink VLAN (the last two save automatically) while the port is down, the disabled state is **saved permanently**. Enable the port before saving the configuration.

NOTE

If the port is re-enabled directly over CLI (not from the NMS), the NMS notices from polling about 10 minutes later and sends a recovery notification.

Per-port ONU list

Open it from a port card, from Port Detail (**View ONU List**), or from a search result. Page title: **ONU Slot X Port Y**.

Header: port navigation (left/right arrows or dropdown), **GPON Port & ONU**, **Bulk TR069**, and **Refresh ONU** (reread this port over SNMP).

Table columns:

COLUMN	CONTENT
ONU	Subscriber name and interface (for example gpon-onu_1/2/1:15)
Serial	ONU serial number
ODP	Dropdown to link the ONU to an ODP
Type	ONU type on the OLT
ONU RX	Colour-coded RX power (green/yellow/red)
Status	Phase: Working/Online, LOS, Dying Gasp (power loss), Offline
Admin	active or disabled
Last Down	Last reason the ONU went down
Actions	Action icons (see below)

Filters: search box (interface, serial, name, type), [All Phases](#), [All Admin](#), ODP filter, and [Reset filter](#).

Links from alarms, notifications, or ONU Monitoring open this page and highlight the target ONU row directly.

Per-ONU actions

ICON	PURPOSE	MODELS	NOTES
Edit ONU info	Rename the ONU and change its description	C300/C320	Uses SNMP SET; needs the Write Community
ONU detail (CLI)	Full detail and RX trend	All	See ONU Detail
Configure ONU (CLI)	Edit the running configuration	C300/C320 (C600 view only)	See Configure ONU
Add to Map	Place a map pin	All	Paste a Google Maps link or pick on the map
Enable/Disable	Activate/deactivate the ONU	C300/C320	Uses SNMP SET; the subscriber is cut off while disabled
Reboot	Restart the ONU	All	Subscriber offline for 30–60 seconds
Delete	Remove the ONU registration (no onu)	All	Permanent; the ONU must be registered again

Bulk actions: tick several ONUs (or [Select all](#)), then:

- [Delete selected](#): deletes all selected ONUs in one CLI session and reports the result per ONU.
- [Copy to another port](#): see Copying ONUs to another port.

CAUTION

Reboot, delete, enable/disable, ONU copy, and bulk TR069 do **not** save the configuration automatically. Run Save Config if the change must survive an OLT reboot.

ONU Detail

[ONU detail \(CLI\)](#) reads data live from the OLT over Telnet.

- Status cards, ONU RX power, distance, and online duration.
- Optics: RX/TX, upstream/downstream attenuation, and a safe zone indicator (Healthy/Warning/Critical).
- Identity, status, last event, and [All Fields](#).
- RX Power trend** over 24 hours, 7 days, or 30 days from polling history. Useful for spotting attenuation that worsens slowly.

On the C600, the OLT CLI masks the name and description as [*****](#); the subscriber name still appears in the ONU list because it is read over SNMP.

Configure ONU

Configure ONU (CLI) reads the ONU running-config from the OLT and shows it by section: name, T-CONT, GEM port, service port, ONU service, UNI VLAN, TR069/ACS, WAN IP, WAN configuration, and Remote ONT (security-mgmt).

Two ways to change it:

- **Per section:** pick a section, Add/Change/Delete, then click **OK · send to OLT**. The configuration is reread afterwards.
- **Full editor:** change several sections at once, review the change script preview, then click **Apply to OLT (CLI)**.

Every change is recorded in the Registration History.

CAUTION

Changing VLANs, service ports, or WAN settings disconnects the subscriber for about 5–10 seconds. Changes are not permanent until you run Save Config.

ONUs with an onu-profile (C300): rows locked by the profile show a padlock (the OLT rejects them with code 64007). **Detach profile** runs `no onu N profile` and rewrites the services; the subscriber is offline for a few seconds.

C600: this page is **view only** ("Read only" label).

Unconfigured ONUs and registration

Finding new ONUs

1. Open **Unconfigured** and choose an OLT (only ZTE OLTs are listed).
2. Click **Refresh Discovery** to read ONUs that are not registered yet.
3. The table shows the Serial, Port, and **First Seen** time (a **New** badge for less than 24 hours).
4. Click **Register ONU** on the row, or **Bind to an existing ONU** to replace a broken ONU (see Bind ONU).

C300/C320 registration form — Simple mode

The right-hand panel shows a **CLI script preview** that updates as you edit the form.

- **ONU identity:** Serial Number, Slot, Port, ONU ID (suggested from free slots), Customer Name.
- **Description (optional):** up to 80 characters. When empty, the NMS uses the automatic `{ONU ID}$$Name$$` format.
- **ODP (optional):** only ODPs on the selected port. Linked after the registration is executed successfully.
- **GPON configuration:** ONU Type, TCONT Profile, VLAN ID, VLAN Profile, Service Name, and Service Mapping Mode (VLAN + Priority or Transparent; some firmware only connects with Transparent).
- **WAN mode:** PPPoE, DHCP, Static (IP Profile, IP, prefix), or Bridge (the subscriber's router does the PPPoE).
- **Additional features:** TR069 (ACS URL, username, and password; the password is filled from Settings) and Remote ONT (ONU web access from the WAN: ID, forward/discard mode, protocol).

Advanced mode (C300/C320)

For multi-service ONUs (for example internet plus hotspot). The header holds SN, slot, port, ONU ID, ONU type, and ODP. The editor below builds T-CONTs, GEM ports, service ports, services, WAN IPs, TR069, and Remote ONT row by row, plus a Description field (empty = same as Name).

C600 registration form (TR069 model)

The C600 uses a SmartOLT-style service layout: two services (internet and management), in-band management IP, and ACS.

- Identity: Serial, Slot, Port, ONU ID, ONU Type, Customer Name, Zone, Description (empty = zone, or the name when the zone is empty), and ODP.
- Services: internet VLAN and its TCONT profile, management/TR069 VLAN and its TCONT profile, and the egress traffic policy.
- Management: management IP (must be unique), subnet, gateway, priority, host, and ACS.
Auto mgmt-IP scans the IPs already used on the OLT and fills in the next free one.
- Tick ONU web access (security-mgmt) if needed.

The C600 does **not** support PPPoE/DHCP/Static WAN modes.

Generate or Execute

- **Generate script only**: stores the script in the Registration History (status "Not executed") without touching the OLT.
- **Execute to OLT**: after confirmation, the script is sent over Telnet. You are taken to the Registration History with the result.

CAUTION

C300/C320 registration scripts do not save automatically. Run Save Config after a batch of registrations. C600 scripts already end with **write**.

Registration History

Open it from OLT detail → **Registrations** or from Unconfigured → **Registration History**. It lists the last 50 registrations with status Not executed, Registered, Config updated, Failed, or Config change failed.

Actions: **Execute to OLT** / **Retry execution**, **Delete script** (not for registered entries), **View script**, and **Logs** (CLI output).

Bind ONU (replacing a broken ONU)

Bind ONU swaps the old ONU serial for a new ONU in the same slot, so the subscriber's whole configuration (VLAN, PPPoE, ODP, map pin) carries over without re-registration. C300/C320 only.

1. Connect the replacement ONU on the subscriber's line, open **Unconfigured**, and click **Refresh Discovery**.
2. On the new ONU row, click **Bind to an existing ONU**.
3. Choose the old ONU from the ONUs on the **same port** (offline ONUs are listed first), then click **Next**.
4. Read the warnings. Red means the old ONU is still online. Only the SN changes; WiFi/web settings on the old ONU do not carry over, and the ACS treats it as a new device.
5. Keep **Save the OLT config (write) afterwards** ticked (default). Without it, the old SN comes back when the OLT reboots.

6. Click **Bind ONU**.

The server rejects an SN that is not detected on the same port. The action is recorded in the **Audit Logs**.

Copying ONUs to another port

Copies ONU configuration from one port to another, for example when moving subscribers to a new PON. C300/C320 only.

1. In the source port's ONU list, tick the ONUs to copy.
2. Click **Copy to another port** and enter the destination Slot and Port.
3. Choose **Generate script** (script only), or tick direct execution and click **Copy & execute**.
4. The job runs in the background with progress; you may close the dialog. Results are stored in the Registration History.

CAUTION

ONU numbers on the destination port are allocated automatically, so **Refresh ONU** the destination port first. GPON serials must be unique on the OLT: move or delete the ONUs from the source port before a real execution.

Bulk TR069

Enables TR069/ACS on **every ONU of one port** at once. C300/C320 only. Requires a configured ACS address.

1. In a port's ONU list, click **Bulk TR069**.
2. Click **Scan (Dry-run)**. The NMS reads each ONU's configuration without writing anything.
3. Review the summary: how many will be enabled, how many are already enabled (skipped), and how many could not be read.
4. Click **Execute to OLT** to write. You can **Rescan** at any time.

ONUs whose TR069 already points to the same ACS are skipped, so it is safe to run repeatedly.

Profile catalogue

The **Profile** icon on an OLT row opens the catalogue of ONU Types, T-CONT Profiles, VLAN Profiles, and IP Profiles used by the registration dropdowns.

- **Sync From OLT** reads the profiles that exist on the OLT.
- **Add** / Edit: tick **Execute CLI** if the profile must also be created on the OLT; without it the profile is stored in the NMS only.
- **Inactive** profiles are hidden from the registration form.
- Delete: choose **Delete from local cache** or **Delete from OLT + cache**.

TIP

CLI results when adding or deleting profiles are not re-checked. After changing profiles on the OLT, click **Sync From OLT** to confirm the outcome.

Configuration backup

OLT detail → [Config Backup](#) (ZTE only).

- **Automatic daily backup:** turn the switch on. The backup runs at 02:30 and only stores a new version when the configuration changed.
- **Backup now:** click [Backup](#) for a manual backup (a few seconds).
- **Compare versions:** pick an old version (A) and a new version (B), then [Compare](#) to see the changed lines.
- Per row: [View content](#) and [Download .txt](#).

Anyone who manages the OLT may create backups and set the daily switch. **Viewing, downloading, and comparing** backup contents is limited to admins/operators or the partner who owns the OLT, because the running-config contains credentials.

ZTE support matrix

FEATURE	C300 / C320	C600
Add OLT, Test SNMP, polling, Save Config, Telnet	Yes	Yes
OLT detail, chassis, Port Detail (status, optics, traffic)	Yes	Yes
PON port description edit, uplink VLANs	Yes	Yes
Disable/Enable PON port	Yes	No
ONU list, filters, ODP column, map	Yes	Yes
Rename ONU, Enable/Disable ONU	Yes (SNMP)	No
Reboot ONU, Delete ONU	Yes	Yes
ONU Detail + RX trend	Yes	Yes (name masked in CLI)
Configure ONU	Yes	View only
Unconfigured	Yes	Yes
Simple/Advanced registration (PPPoE/DHCP/Static/Bridge)	Yes	No
TR069-model registration (automatic management IP)	—	Yes
Description and ODP during registration	Yes	Yes
Bind ONU, Copy ONUs to another port, Bulk TR069	Yes	No
Profile catalogue, configuration backup	Yes	Yes

06 C-Data OLTs

Adding a C-Data OLT

1. **SmartOLT** → **OLT C-Data** tab → **Add OLT**.
2. Pick the correct **OLT Family**: C-Data EPON (17409) or C-Data GPON (34592).
3. Fill in the IP, SNMP (Read Community required), and Telnet CLI.
4. Click **Save OLT**. The NMS scans immediately and shows "OLT added. Initial scan: N ONUs."

CAUTION

C-Data GPON FD160x OLTs report the same sysObjectID as EPON (17409). That is why **the Family chosen in the form decides**. Choosing the wrong one makes a GPON OLT behave as EPON. An OLT name containing the model (for example **FD1608**) is recognised as GPON automatically.

When Telnet is needed: the ONU inventory is still read over SNMP without Telnet, but every write action (rename, reboot, enable/disable, delete, VLANs, save configuration) and the extra C-Data GPON data (RX from CLI, admin status) require Telnet.

OLT detail and front panel

Click **Details**. Header buttons: **Back**, **Edit**, **VLAN** (V3 firmware only), **EPON Ports** / **GPON Ports**, and **Scan ONUs**.

- Summary: Total ONUs, Online, Offline, and the number of PON ports up.
- System information: description, system name, uptime, firmware (FlashV3.x or Legacy/SNMP), model, serial, HW/SW versions, and last scan time.
- **Front panel**: physical port picture with real-time status (Up, Down, Shutdown, copper RJ45, fibre SFP). The number under each PON port is online/total ONUs.
 - V3 firmware: any port (PON, GE, XGE) can be clicked to open **Port Detail**.
 - Older firmware: PON ports open the ONU list.

PON ports and port labels

EPON Ports / **GPON Ports** shows a card per port: status, **NMS label**, and an online/total ONU bar.

A **port label** is a note stored in the NMS (not written to the OLT), because the C-Data/HIOSO port description commands are not verified.

1. Open a port's ONU list.
2. In the header, click the pencil icon **Add port label**.
3. Enter a label (up to 64 characters, for example "East route / Griya estate") and press Enter. Clear it to remove.

Labels survive polling and also appear in the Android app.

Per-port ONU list

Title: {OLT} · EPON/GPON 0/slot/port.

Columns: ONU (interface and type), Serial/MAC (EPON ONUs use the MAC), Name, ODP, Status (Online/Offline; C-Data GPON also shows LOS/dying-gasp), Rx, and Actions.

- Search: serial, name, interface, or MAC. The ODP filter appears when the port has ODPs.
- There are no phase/admin filters as on ZTE.
- [Refresh](#) scans this port only.

C-Data ONU actions

Action icons are shown to admins, operators, and partners. All of them require Telnet.

ACTION	NOTES
Rename	Up to 128 characters; clear to remove the name
Reboot	Subscriber offline for about 30–60 seconds
Enable/Disable	Deactivates the ONU without deleting its registration
Delete	Removes the ONU registration; the ONU must be registered again on the OLT
Remote ONT	Opens/closes ONT web access from the WAN (see below)

NOTE

C-Data EPON does not report admin status over SNMP. The enable/disable state you see is only the result of the last NMS action and returns to "active" after the next scan. C-Data GPON with Telnet reads the real admin status.

Remote ONT (C-Data GPON FlashV3)

Opens the ONT web page from the WAN side, for example so a technician can reach a subscriber's ONT remotely. It also works for ZTE-branded ONTs.

1. Click the globe icon on the ONU row.
2. In [Remote ONT \(web access\)](#), click [Enable](#) or [Disable](#). It takes effect instantly, without a reboot.
3. Run **Save Config** so the rule survives an OLT reboot.

CAUTION

The rule has no source-IP filter, so the ONT web page is exposed to the internet. Close it again when you are done. Not available on FD1601S/FD1602S, C-Data EPON, or HiOSO.

VLANs and Port Detail (V3 firmware)

Available on C-Data OLTs with V3 firmware (proven on FD1608S, FD1601S, and EPON FD1304E). Data is read live from the CLI; if that fails, the NMS shows the last successful reading (up to 7 days old).

VLAN page

OLT detail → [VLAN](#). Shows total VLANs, VLANs on the uplink, VLANs not on the uplink, and L3 VLANs.

Adding a VLAN:

1. Enter the VLAN ID (1–4094) and an optional description (one word: letters, digits, dot, [-](#) or [_](#)).
2. Tick the ports to tag: uplinks and/or PON ports (up to 24 ports).

3. Click [Create & tag](#) and confirm.

4. Click [Save Config](#) to make it permanent. The NMS does **not** save automatically.

GPON versus EPON:

- **GPON:** PON ports join every new VLAN automatically. Tagging the uplink is enough.
- **EPON:** PON ports and some uplinks run in Hybrid mode. **EPON ports must be ticked manually.**

Port Detail

Click a port on the front panel or in the VLAN list.

- Status: admin, link, speed, optical module, VLAN mode, native VLAN, max frame, flow control, link uptime, last down reason, OLT description, and NMS label.
- Optics (DDM): RX/TX power (coloured when outside thresholds), vendor, PN/SN, wavelength, bias, temperature, voltage. RX on a PON port is the combined power of all ONUs, not one subscriber's attenuation.
- Traffic: RX/TX rate, packets per second, utilisation, and counters since reset.
- Tagged/untagged VLANs, with a [Tag VLAN](#) field for Trunk/Hybrid ports.

Who may write (create VLANs, tag, Save Config on this page): admins/operators or the partner who owns the OLT, with Telnet. Every write is recorded in the [Audit Logs](#).

C-Data online status and RX

TYPE	ONLINE STATUS	RX SOURCE
C-Data EPON	SNMP, Online/Offline only	SNMP
C-Data GPON FlashV3 (with Telnet)	SNMP, with LOS/dying-gasp cause	CLI (SNMP fallback)
C-Data GPON FD1601S/FD1602S	SNMP	SNMP

Offline ONUs show no RX. The RX trend chart is available for ZTE only.

07 HiOSO OLTs

Supported variants

VARIANT	PORTS	NOTES
HA7304	4 EPON ports	Panel layout: PON 1–4, G1/G2 SFP, G3/G4 RJ45
HA7302 (CSM/CST)	2 EPON ports	SNMP does not separate the ports, so the NMS shows one combined port <code>epon 0/1/1</code> holding every ONU (up to 128)

Adding a HiOSO OLT works like any other vendor: `OLT HiOSO` tab → `Add OLT`. The family is locked to "HiOSO EPON (25355)" and CLI is Telnet only. The HA7302 variant is recognised automatically from its firmware during scanning.

OLT detail and ports

- System information shows the firmware string, for example `v7.76/HA7302CSM/...`.
- Front panel:** PON port status is derived from the number of online ONUs (any online = Up, all offline = Down); G uplink status comes from SNMP. Click a PON port to open its ONU list.
- Port labels, the ONU list, the ODP filter, and the map work the same as for C-Data.

HiOSO ONU actions

ACTION	HA7304	HA7302
Rename	CLI; up to 32 characters, letters/digits/ <code>_ - .</code> only (spaces become <code>_</code>), cannot be empty	SNMP SET; up to 32 characters; needs the SNMP Write Community
Reboot	Yes	Yes
Enable/Disable	Yes	Yes (blocks/allows ONU authentication)
Delete	Yes	Yes

NOTE

The rename dialog's helper text still says "max 128, leave empty to clear". For HiOSO, the limits in the table above apply.

HiOSO online status and RX

- Online status comes from link state**, not from RX. Some ONUs report no RX ("na") even while online and passing traffic. The Rx column can therefore be empty for an Online ONU; that is normal.
- The last RX value is kept for at most 2 polls, then cleared so no frozen figure is shown.
- Anti-flap:** an ONU that was online is only marked offline after being read as down in 2 consecutive polls. One bad reading does not make the dashboard flicker.
- PON port status is derived: a port counts as down when all its ONUs are offline.

C-Data and HiOSO support matrix

FEATURE	C-DATA EPON	C-DATA GPON FLASHV3	C-DATA GPON FD1601S/FD1602S	HIOSO HA7304	HIOSO HA7302
Add OLT, Test SNMP, polling, scan	Yes	Yes	Yes	Yes	Yes (1 combined port)
Front panel	Yes	Yes → port detail	FD1601S → port detail	Yes → ONU list	Yes → ONU list
NMS port label	Yes	Yes	Yes	Yes	Yes
Rename ONU	CLI, ≤128	CLI, ≤128	CLI, ≤128	CLI, ≤32	SNMP, ≤32
Reboot, Enable/Disable, Delete ONU	Yes	Yes	Yes	Yes	Yes
ONU admin status readable	No	Yes (Telnet)	Yes (Telnet)	No	No
Remote ONT	No	Yes	No	No	No
VLAN + Port Detail	V3 firmware only	Yes	FD1601S: Yes	No	No
Save Config	Yes	Yes	Yes	Yes	Yes
Separate LOS / dying gasp alarms	No (offline only)	Yes	Yes	No	No
ONU registration, Configure ONU, config backup, Bulk TR069, ONU Detail	No	No	No	No	No

08 ONU Map and ODPs

ONU Map

ONU Map shows subscriber ONUs and ODPs from every OLT on one map.

- **ONU pins** are **green** (online) or **red** (offline, LOS, or power loss; pulsing). The colour does not indicate attenuation; the RX value is on the pin card.
- **ODP pins** are yellow by default (or their port colour) with the number of ONUs.
- **Cable lines** animate from ODP to ONU, green/red following the ONU status.
- **Layers:** Google Streets, Satellite, Hybrid, Terrain, with OpenStreetMap as a fallback. Google tiles are used without an API key; if they are blocked, switch to OpenStreetMap.

At low zoom with more than 350 pins, pins are drawn as dots to keep the map responsive.

Adding a pin from the map

1. Click **Add Pin**. A note says add mode is active.
2. Click the subscriber's location on the map.
3. In the dialog, choose **ONU** or **ODP**.
4. ONU: search for the ONU or pick OLT → Port → ONU, optionally fill in the customer name, phone, address, and notes, then **Save Pin**.
5. ODP: enter the ODP Name and (optionally) the PON port, then **Save ODP**.

Adding a pin from the port page

In a per-port ONU list, click **Add to Map**:

- **Paste a Google Maps link** (from the technician's Maps app), then click **Place**. The pin is placed at those coordinates.
- Or choose **Open Map & place pin** to pick the location on the map.

ONU pin card

Click a pin to open its card:

- **Unlock** / **Lock**: new pins are always locked. Once unlocked, the pin can be dragged and its position is **saved automatically** when released. **Lock** only closes it again.
- **Edit Name**: the ONU name is written straight to the OLT.
- **Reboot** the ONU, **ONU Detail** (ZTE), **Port** (open the port page), and a Google Maps link.
- **Delete Pin**: removes only the pin; the ONU on the OLT is unaffected.

ODPs

ODP lists every ODP (field splitter) with its connected ONUs. The rules: one ONU connects to one ODP, and one ODP belongs to one PON port.

Adding or editing an ODP

1. Click **Add ODP** (or the edit icon on an ODP row).

2. Enter the Name, OLT, and Slot/PON Port (optional; filled automatically when the first ONU is linked).
3. Paste a [Google Maps link](#) and click [Get coordinates](#), or type the latitude/longitude manually.
4. Add notes if needed and save.

When editing, an ODP can be moved to another OLT, slot, or port. If some ONUs no longer match the new port, the NMS warns how many will be unlinked before saving.

Managing ONUs in an ODP

Click [Manage ONUs](#) on an ODP row:

- **ONUs in this ODP:** click [Remove from ODP](#) to unlink.
- **Add ONU:** pick an ONU, then [Add to this ODP](#). A "now in another ODP" label means the ONU will be moved from its current ODP.

ONUs can also be linked from the **ODP** column in the per-port ONU list, or during ZTE ONU registration.

ODP photo

Each ODP can have one documentation photo (JPG/PNG/WebP, up to 12 MB; converted to WebP automatically). Uploading a new photo replaces the old one. Photos are stored privately and only visible to signed-in users. Technicians can take the photo straight from the Android app.

ODP colour per PON port

Colours group ODPs per port on the map.

1. Open the ODP card on the map or the ODP row, then click [Colour](#).
2. Pick one of 16 colours, [Random](#), or [Default](#).
3. The "Apply to every ODP on port ..." switch is on by default, so every ODP on that port changes colour.

New ODPs, or ODPs moved to a port, inherit that port's colour automatically. Green and red are deliberately unavailable because they represent ONU status.

Deleting an ODP

DANGER

Deleting an ODP is **permanent**: the links to all its ONUs are removed too, and the action is **not recorded in the Audit Logs**. ONUs on the OLT are unaffected. Recovery is only possible from the daily database backup by a server admin.

09 Alarms and Notifications

Alarm types

TYPE	SEVERITY	WHEN IT APPEARS
OLT unreachable	critical	The OLT cannot be read over SNMP
PON port down	critical	A PON port goes from up to down
PON port disabled by admin	major	A port is disabled from the NMS; sent immediately
ODP down (all ONUs offline)	major	Every ONU (at least 2) of one ODP goes offline together
Loss of Signal (LOS)	major	The ONU loses optical signal (cut or bent fibre)
Power loss (dying gasp)	minor	The ONU reports losing mains power
ONU offline	minor	The ONU is offline without a more specific cause
High RX attenuation	warning	$RX \leq -28$ dBm or ≥ -8 dBm; considered recovered between -26 and -10 dBm

Separate LOS and dying gasp alarms are only available for ZTE and C-Data GPON. C-Data EPON and HiOSO always report "ONU offline".

Alarm lifecycle

- **Pending:** the fault is detected for the first time. It is not sent and not shown on screen.
- **Active:** the fault is still present at the next poll (about 10 minutes later), and the notification is sent. This delay (anti-flap) prevents false notifications from momentary glitches. If the fault clears before confirmation, the pending entry is deleted silently without any message.
- **Cleared:** the fault recovers. A recovery notification is sent if enabled in Settings.

Alarms are only raised on a **change** from healthy to faulty. An ONU that has been down for a long time does not raise a new alarm at every poll.

Root-cause correlation

So that recipients are not flooded with hundreds of messages when one cable breaks, the NMS sends only the **parent** alarm:

- OLT unreachable → ports and ONUs of that OLT are not evaluated.
- PON port down → new ONU alarms on that port are held back. The port-down message states how many ONUs are affected.
- Every ONU of one ODP offline → a single "ODP down" alarm.
- Some ONUs of one ODP down → summarised into one message listing the subscribers (a separate option in Settings).
- Port disabled by admin → port-down, ONU, and ODP alarms for that port are held back until the port is enabled.

When the parent recovers but some ONUs are still down, those ONUs are reported as faults of their own.

Alarms page

Alarms has the tabs **Active**, **Cleared**, and **All**.

- Filters: search box (serial, message, type, OLT), Severity, OLT, Scope (olt/port/odp/onu), and Type. Click **Apply** or **Reset**.
- Columns: Severity, Type, OLT/Target, Message, Status, last seen and since, and the customer name.
- Click an ONU alarm row to open the related ONU (ZTE OLTs).

NOTE

"ONU offline" and "Power loss" alarms that you have **marked as read** disappear from the Active tab until they recover, then show under Cleared. LOS, attenuation, port, ODP, and OLT alarms stay visible until they actually recover.

Notification bell

The red badge on the header bell shows the number of unread active alarms.

- The dropdown holds 8 alarms: unread ones first, then important alarms that are still active. It refreshes every 60 seconds while open.
- Click an item to open the related ONU, port, or OLT directly. If the ONU moved to another port, its new position opens. If its old position is now used by a different subscriber, the NMS refuses to open it so you do not manage the wrong ONU.
- The tick icon is **Mark as read**. There is also **Mark all as read**.

Alarm settings

Settings → **Alarm** tab (admin). These settings apply to the main Telegram bot and app push. Alarms are always recorded in the history regardless; only delivery is controlled here.

OPTION	PURPOSE
Confirm over 2 polls before sending	Anti-flap (recommended). Turn off for real-time mode
Send only the parent alarm when a PON port / ODP is down	Root-cause correlation
Summarise several ONUs of one ODP into one message	One message per ODP for partial outages
Minimum severity	Warning (all), Minor and above, Major and above, or Critical only
Send when a new alarm appears / when it clears	Notification triggers. Recovery messages (including "port enabled again") are off by default on a new installation
Alarm types to send	Tick the types you want delivered

Main Telegram bot

Settings → **Telegram Bot** tab (admin).

1. Create a bot with @BotFather in Telegram and copy its token.

2. Fill in the **Bot Token** and **Chat ID** (several allowed, comma-separated; group IDs start with a minus sign).
3. Turn on **Enable notifications**, then click **Send Test**.
4. Optional: turn on **Enable bot commands** and click **Register Webhook** so the bot answers commands.

Available bot commands include `/menu`, `/status`, `/olt`, `/los`, `/redaman`, `/cari`, `/alarm`, `/onu`, `/prov`, `/uncfg`, `/refresh`, and `/id`, plus a button menu. Rebooting an ONU from the bot always asks for confirmation.

Partner Telegram bot

Partners have **My Telegram Bot** with their own bot, chat IDs, and **own filters** (minimum severity, triggers, and alarm types). A partner bot only serves OLTs owned by or assigned to that partner, and follows the partner's per-OLT alarm switches.

Android app push notifications

Settings → **Mobile Notifications** tab (admin).

- **Enable alarm push to mobile**: a warning appears if the Firebase credentials are not installed on the server.
- **Send a Manual Notification**: enter a title and message, then **Send to All Devices** (for example a maintenance announcement).
- **Registered Mobile Devices**: phones that are signed in; click **Revoke** to end a device session.

Push stops automatically when a user signs out of the app.

10 Reports and Audit Logs

Reports

Reports produces downloadable reports. Every role can use it; the content follows the user's OLT scope.

TYPE	CONTENT
ONU Inventory & RX Power	ONU list with status and attenuation
OLT Status	Condition of every OLT
Alarm History	Alarms within a time range
ONU Provisioning	ONU registration history

1. Choose the report type.
2. Set the filters: time range (for alarms and provisioning: 24 hours, 7 days, 30 days, or all time), OLT, PON port, attenuation (Normal ≥ -25 , Warning < -25 , Critical < -28 dBm), and status.
3. Click CSV (opens directly in Excel) or PDF (A4 landscape).

Audit Logs

Audit Logs (admin) records who did what and when.

- Filters: search box (description, user, IP), Event, User, and date range. Click **View details** for the full record.
- Recorded events include: records created, updated, or deleted; sign-in, sign-out, and failed sign-in; Telnet terminal sessions; Bind ONU; port disable/enable; VLAN writes.

NOTE

ODP deletions are **not** recorded in the Audit Logs.

11 Users and Permissions

Managing users

Users (admin):

1. Click the add button, then enter the name, email, password, and role.
2. For operators and partners, tick the OLTs they may access under **OLT Assignment**.
3. Save, then give the user the application address and their password.

Existing users can be edited (name, email, role, password, assignment) or deleted. An admin cannot delete themselves or the last admin.

OLT assignment

- **Operator:** with no assignment, the operator can access every OLT. With an assignment, only the ticked OLTs.
- **Partner:** only the ticked OLTs, plus their own private OLTs.
- Deleting a partner returns their private OLTs to the shared pool.

Permission matrix

CAPABILITY	ADMIN	OPERATOR	PARTNER	DEMO
View Dashboard, Monitoring, Map, ODP, Alarms, Reports	Yes	Yes	Yes (their OLTs)	Demo data
ONU actions (reboot, rename, delete, registration)	Yes	Yes	Yes (their OLTs)	No
Map pins, ODPs, photos, colours	Yes	Yes	Yes	No
Add OLT	Yes	Yes	Yes (becomes private)	No
Delete OLT	Yes	Yes	Own only	No
Change OLT connection, Test SNMP, Telnet, backup contents	Yes	Yes	Own only	No
Disable/Enable PON port	Yes	No	Own only	No
Per-OLT alarm switch	Yes (global)	Yes (global)	Yes (their own)	No
Users	Yes	No	No	No
Audit Logs, Settings, API tokens	Yes	No	No	No
My Telegram Bot	No	No	Yes	No

NOTE

Demo accounts only see demo-labelled data. Every change is rejected, including marking notifications as read and changing the language; only signing out and changing the theme are allowed.

12 Settings

Settings is for admins only.

TAB	CONTENT
General	Logo and Application Name, Application Version, System Information, and the Android App card with Download APK
ACS / TR069	ACS endpoint (URL, username, password) planted in ONUs during TR069 registration and used by Bulk TR069
Alarm	Alarm delivery policy (see the Alarms chapter)
Telegram Bot	Main bot connection and command webhook
Mobile Notifications	Firebase push, manual notifications, registered devices
API & Tokens	REST API tokens for integrations

13 Android App

Installing the app

The Android app is built from the `mobile/` folder and **must** be given the server address at build time:

```
API_BASE_URL=https://nms.your-domain.com/api/v1 bash bin/build-apk.sh
```

The build needs at least 8 GB RAM, 15 GB disk, JDK 17, Flutter, and the Android SDK (see `docs/BUILD_APK.md`). The result is copied to `public/downloads/kusumavision-nms.apk`, so it can be downloaded from `Settings` → `General` → `Download APK` or from `https://<server>/downloads/kusumavision-nms.apk`. An APK built without `API_BASE_URL` only shows a message that the server address is not configured.

Sign in with your NMS email and password.

App tabs

TAB	CONTENT
Dashboard	Network summary; magnifier icon for search
OLT	OLT → port → ONU → ONU detail: reboot, rename, delete ONU; Unconfigured and basic-mode ZTE ONU registration (with optional ODP)
ODP	ODP list and search, per-OLT filter, ODP detail with its ONUs, colour change, and photos (Camera/Gallery/Delete)
Map	ONU and ODP map (view only), with Map/Satellite toggle
Account	Account info, Alarms card, notification test, theme (System/Light/Dark), and Sign out

Push notifications: tap an alarm notification to open the alarm or the related ONU directly.

Web only: creating/editing/deleting ODPs, adding and dragging map pins, and enabling/disabling ONUs and C-Data Remote ONT.

14 REST API

The NMS offers REST API v1 for integration with other systems (for example a billing application or a third-party dashboard).

- **Base address:** `/api/v1`. Every request uses the `Authorization: Bearer <token>` header. Limit: 120 requests per minute per token.
- **Creating a token:** `Settings` → `API & Tokens` → enter a token name → `Create Token`. The token is **shown only once**, so copy it right away. Tokens are valid for 30 days by default and can be revoked from the same table.
- **Read endpoints:** summary, OLTs, ONUs, ONU detail, search, alarms (with a target location for deep links), ODPs, map, unconfigured.
- **Write endpoints:** ONU registration (ZTE), reboot, rename, ONU delete, and ODP colour and photo. Not available to demo accounts.
- **Without a token:** `GET /api/v1/public/status` returns aggregate counts only, safe to embed on a website.

15 Safe Practice

When configuration is saved permanently

OLT configuration is lost on reboot unless it has been saved (**write** / **save**). This table shows which actions save automatically.

ACTION	SAVED AUTOMATICALLY?
ZTE C300/C320 ONU registration	No
ZTE C600 ONU registration	Yes
Bind ONU	Yes (ticked by default, can be turned off)
ZTE port description edit, ZTE uplink VLAN	Yes
PON port disable/enable	No
ONU reboot, delete, enable/disable	No
Configure ONU, ONU copy, Bulk TR069	No
Profiles with Execute CLI	No
C-Data VLANs and port tagging	No (press Save Config)
C-Data Remote ONT	No (press Save Config)

Recommended habits:

1. After a registration session or configuration change, run **Save Config** on that OLT.
2. **Do not** run Save Config while a port is intentionally disabled for the time being. Enable the port first.
3. For ZTE, turn on **Automatic daily backup** so every configuration change has a version trail.

Actions that interrupt subscribers

ACTION	IMPACT
Disable PON port	Every ONU on the port is cut off until enabled
Disable ONU	One subscriber is cut off until enabled
Reboot ONU	30–60 seconds
Configure ONU (VLAN, service port, WAN)	5–10 seconds
Detach onu-profile	A few seconds
WiFi change through the ACS	Subscriber devices must reconnect
Delete ONU	Subscriber is cut off until registered again

Other tips

- Always **Refresh** port data before a large action (ONU copy, Bind ONU), because tables show the last poll.

- Use the **per-OLT alarm switch** during scheduled maintenance to avoid a notification flood, and turn it back on afterwards.
- The Telnet terminal gives full CLI access. Use it only when the NMS features are not enough.

16 Troubleshooting

SYMPTOM	COMMON CAUSE	WHAT TO DO
Test SNMP fails	Wrong IP/port or community, or a firewall blocking UDP/161	Check the read community on the OLT and the firewall/SNMP ACL
Write action fails as "not supported"	CLI Transport is not Telnet, credentials are empty, or the feature is not available for that model	Fill in Telnet CLI under Edit OLT; check the support matrix
ZTE ONU rename/enable fails	SNMP Write Community empty or wrong	Fill in the SNMP Write Community under Edit OLT
ONU data does not change	Still showing the last poll	Click Refresh ONU or Scan ONUs
HiOSO ONU Online but RX empty	The OLT does not report RX for some ONUs	Normal; online status comes from link state
Registration lost after an OLT reboot	Configuration not saved	Run Save Config after C300/C320 registrations
No Telegram notifications	OLT alarm switch off, alarm type not ticked, or minimum severity too high	Check the per-OLT switch and the Alarm tab in Settings; click Send Test
Clicking an alarm does not open the ONU	The ONU was deleted, or its position is used by another subscriber	Open the Alarms page and search by serial
Sign-in fails with error 419	App behind a proxy/Cloudflare without <code>TRUSTED_PROXIES</code>	Set <code>TRUSTED_PROXIES</code> , then run <code>php artisan config:cache</code>
Password reset email never arrives	<code>MAIL_MAILER=Log</code>	Configure SMTP in <code>.env</code>